

Social Simon Effect between Two Adjacent Avatars in VR

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ABSTRACT

Joint actions are increasingly taking place in virtual reality (VR), as social VR and other forms of VR applications have become more prevalent. It has been suggested that when engaging in joint actions in real world, each individual achieves unconscious coordination with the other by automatically activating representations of the other's behavior in their motor system. In particular, the conditions under which such implicit cooperation with others occurs and how it can be facilitated have been investigated by examining the Social Simon Effect (SSE). However, collaborative partners are not always perceived in the same way between the real world and virtual reality. In particular, the visual representation of the avatar is thought to affect the perception of others' presence and actions. This study looked into (1) whether SSE can occur during joint actions of avatars in VR; and (2) how visual representation of the co-actor's avatar affects SSE. Results showed that SSE was induced when the co-actor's avatar was displayed in full-body or was entirely transparent, but it was absent (or weak) when only the two hands were visible.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Collaborative interaction

1 INTRODUCTION

Co-acting with others is a fundamental aspect of human life, which enables us to seamlessly collaborate with others in achieving common goals. Previous studies have indicated that individuals in joint action tend to automatically represent their co-actor's task and integrate this representation into their own action planning, which is called "co-representation". Automatic task co-representation has been classically demonstrated by the Social Simon Effect (SSE, or joint Simon effect), which is a spatial stimulus–response interference effect induced by the presence and actively engagement of a co-actor in a joint single response go/nogo task (the joint Simon task) [3]. In the joint Simon task, two types of stimulus (e.g., red/green dot, or two types of auditory pitch; these stimuli themselves are without any spatial information) are presented at the left or right side of some reference point (e.g., a central fixation cross displayed on a screen). Fig. 1 shows one example of the visual version of the joint Simon task. One participant (represented in red) who is sitting on the right side is responsible for responding to the red dot stimuli by pressing the button in front of them as quickly as possible, regardless of the location of stimuli. If there is a co-actor (represented in green) who is sitting on the left side and designated to respond to the green dot stimuli, the reaction time for the red participant is significantly faster when Stimulus-Response (S-R) is compatible, as opposed to when it's incompatible. Conversely, if the green co-actor is absent, there is no significant difference in the reaction time. This phenomenon is called the social Simon effect (SSE).

The elicitation of SSE has been found to rely on social context; it is not elicited in situations where the co-actor is absent or not

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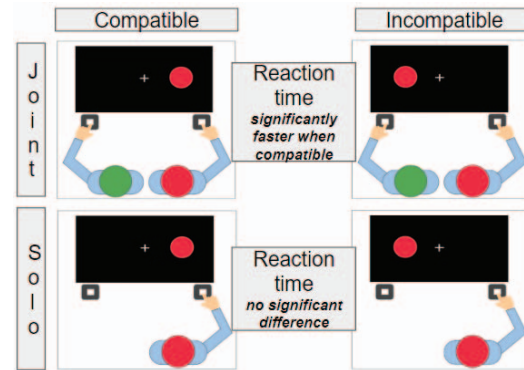


Figure 1: An example of the visual version of the joint Simon task from the perspective of the red participant.

actively involved in the interaction [3]. With social interactions increasingly taking place in VR, one question remains: can SSE be observed during interactions between two avatars in VR, similar to real-world interactions? Because the virtual avatars in VR have limited anthropomorphic properties, the sense of social presence (a sense of being together with social actors) towards avatars is diminished [1], which may negatively impact the manifestation of SSE. Moreover, the users' avatar is not always represented in full-body; avatars with limited visual information such as hand only or fully transparent avatars are frequently used in social VR and VR games. While manipulation of an avatar's appearance has been demonstrated to affect social presence [2], its effect on SSE remains undetermined. In this study, we investigated whether SSE can be observed between two adjacent virtual avatars (participant and co-actor) in VR performing the joint Simon task, and further explored how different visual representations of the co-actor's avatar (full-body, hand, or transparent) influence SSE.

2 EXPERIMENT DESIGN

We conducted a within-subject experiment with four conditions (order counterbalanced): *Solo* (the co-actor did not exist), *Full Avatar* (the co-actor used a full-body avatar of which all body parts were displayed), *Hand Avatar* (the co-actor used a hand avatar of which only the left and right hand were displayed), *Trans Avatar* (the co-actor used a transparent avatar of which none of the body parts was displayed). For the avatars, We used the male avatar (Male_Adult_08) and the female avatar (Female_Adult_05) from the Microsoft Rocketbox Avatar Library. The gender of the participants' avatar matched their biological gender, and the gender of the co-actor's avatar was counterbalanced. In the *Trans Avatar* condition, neither the body parts nor movements of the co-actor's avatar could be visually confirmed, but the participants could discern the effects of these movements (button cap went down and up in the joint Simon task). Movements of the co-actor's avatar were pre-recorded, with responses in the joint Simon task being software-controlled. None of the participants noticed this manipulation during the experiment.

The experimental system was developed using the game engine Unity (version 2021.3.11f1) and ran on a Windows laptop (GAL-

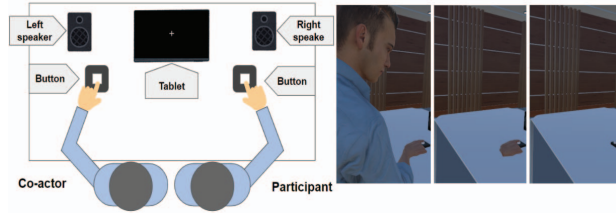


Figure 2: Left: Experimental setup of the audio version joint Simon task in VR. Right: The visual representation of the co-actor's avatars (male) in the *Full Avatar*, *Hand Avatar*, *Trans Avatar* conditions.

LERIA ZL7C-R38H). Twenty-four right-handed participants were recruited (12 males and 12 females, age: 25.708 ± 5.752), and they were paid 2,000 Japanese yen for their 1h15mins participation. During the experiment, participants wore the Meta Quest2 to immerse themselves in VR. Afterwards, as a tutorial, they performed the joint Simon task with 16 trials alone. Then, for each condition, participants performed a cube-carrying task and the joint Simon task. During the cube-carrying task, they carried the virtual cube handed over by their co-actor to the right side of table (or did it alone in the *Solo* condition). This task was designed to make participants to confirm the appearance of the co-actor's avatar and the co-actor's active engagement (or absence of the co-actor). Following the cube-carrying task, the audio version joint Simon task began. Each trail of this task began with a white fixation cross that appeared at the center of the tablet's screen and a warning sound (white noise, 300ms). After 700ms, one of two types of auditory stimulus, Sound A (300Hz pure tone) or Sound B (600Hz pure tone) with a duration of 300ms, was presented from the speaker on either the left or the right side. Participants and their co-actor had to respond exclusively to the assigned auditory stimulus by pressing the button in front of them, no matter which side the auditory stimulus presented. If anyone responded to the stimulus in 1700ms or 1700ms had passed, an empty square was displayed around the fixation cross, which marked the end of one trail. After that, the subsequent trial started after 1000ms. For each condition, there are two sections, with each session consisted of 128 trials (spatial compatibility of S-R relationship (compatible/incompatible) $\times$ sound type (Sound A/Sound B) $\times$ 32 repetitions). After participants finished the joint Simon task in each condition, they filled in a questionnaire from [1] that measured two facets of the social presence: co-presence (the degree to which the individuals feel as if they are in the same space with the other) and perceived attentional engagement (the degree to which the individuals allocates focal attention to the other).

In order to examine the SSE, which was revealed by the compatibility effect (the difference in reaction times between compatible and incompatible S-R), in each experimental condition, we performed a paired t-test. We found a significant difference in reaction times between compatible and incompatible S-R in the *Full Avatar* condition ($t(23) = -2.69$, $p = 0.013$, *Cohen's d* = -0.550) and in the *Trans Avatar* condition ($t(23) = -3.26$, $p = 0.003$, *Cohen's d* = -0.666), such trend was not found in the *Solo* condition ($t(23) = -0.999$, $p = 0.328$, *Cohen's d* = -0.204) or the *Hand Avatar* Condition ($t(23) = -1.96$, $p = 0.062$, *Cohen's d* = -0.400).

Results of one-way repeated measures ANOVA revealed that there was a statistically significant difference in the co-presence score under different conditions ($F(2,46) = 18.431$, $p < .00001$, *generalized η^2* = 0.2). Post-hoc pairwise paired t-test comparisons with a Bonferroni p-value adjustment revealed that the co-presence score in the *Trans Avatar* condition was significantly lower than the *Full Avatar* Condition ($t(23) = 5.30$, $p < .0001$, *Cohen's d* = 1.08) and the *Hand Avatar* Condition ($t(23) = 4.32$, $p < .001$, *Cohen's d* = 0.881). For the attentional engagement

score, although the one-way repeated measures ANOVA revealed a significant effect of experimental condition on the attentional engagement score ($F(2,46) = 3.517$, $p = 0.038$, *generalized η^2* = 0.041), post-hoc pairwise paired t-test showed no significant difference in all condition pairs.

3 DISCUSSION & CONCLUSION

Throughout the experiment, the occurrence of SSE was observed in the *Full Avatar* condition, but it was absent in the *Solo* condition. These results revealed that the full-body avatar, which mirrored the physical movements of the co-actor and was responsible for the complementary portion of the joint Simon task, could be perceived as a socially relevant agent. Participants represented actions of the full-body avatar into their own motor plan, leading to the formation of the task co-representation. SSE was also significant in the *Trans Avatar* condition. Within this condition, although the body parts and body movements (such as pressing and releasing the button in the joint Simon task) of the co-actor's avatar were invisible, a portion of the response actions (like the button cap went up and down) remained perceivable. The observation of another agent's actions can modify actions programmed by the observer and activates response codes in the observer that are functionally equivalent to an actual planned action [3], which may responsible for the occurrence of SSE. Another possible explanation is that in this condition, participants complemented the appearance of the co-actor's avatar through imagination, drawing on their own full-body avatar as a reference. SSE was not significant (or very weak due to the p-value of 0.062 that close to the margin of statistical significance) in the *Hand Avatar* condition. One possible explanation for this attenuated SSE can be that the participants perceived the avatar with only hands as a non-human artificial agent rather than an intentional humanoid agent. The non-human-like appearance of the co-actor's hand avatar might make the participants perceive that they acted in a machine-like manner, thus impeding the action representation process [4]. Regarding social presence, the co-presence significantly declined when all body parts, or all body parts except the hands, became transparent. However, the attentional engagement almost maintained at a similar level. This suggests that neither the co-presence nor the attentional engagement was the decisive factor of SSE. Thus, further research into the relationship between SSE and social presence is required. Taken together, we found that SSE induced when the co-actor was displayed in full-body avatar and transparent avatar, but it was absent (or weak) when they displayed in hand avatar. Results of this study may offer reference for avatar design especially in VR collaboration scenarios.

ACKNOWLEDGMENTS

This work was partially supported by Grant-in-Aid for Scientific Research (S) (19H05661) and JST Moonshot Research & Development Program (JPMJMS2013).

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